

ASSIGNMENT - IX

IX. Application of Laplace Transforms to Differential Equations, Simultaneous Differential Equations :

Solve the following using Laplace Transforms :

1. $\frac{d^2y}{dx^2} + \omega^2 y = 0; y(0) = A; \left(\frac{dy}{dx}\right)_{x=0} = B$

Ans : $y = A \cos \omega x + \frac{B}{\omega} \sin \omega x$

2. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-t}; y(0) = 0; y'(0) = 0$

Ans : $y = e^{-t} - (1+t)e^{-2t}$

3. $y'''(t) + 2y''(t) - y'(t) - 2y(t) = 0; y(0) = y'(0) = 0; y''(0) = 6$

Ans : $y = e^t - 3e^{-t} + 2e^{-2t}$

4. $\frac{d^2y}{dt^2} + \frac{2dy}{dt} - 3y = \sin t; y(0) = 0; y'(0) = 0$

Ans : $y = -\frac{1}{10} \cos t - \frac{1}{5} \sin t - \frac{1}{40} e^{-3t} + \frac{1}{8} e^t$

5. $y'' + 2y' + 5y = e^{-t} \sin t; y(0) = 0; y'(0) = 0$

Ans : $y = \frac{1}{3} e^{-t} (\sin t - \sin 2t)$